Product Specification: Stock Dailies Application

Overview

Background

- Users that perform short-term stock trades (< 1 year) use stock charts to determine when to buy and sell stock
    - The objective behind using charts is to identify the volume of money being moved and get ahead of momentum caused by larger institutions moving money
- Buy and stop limits to sell stock have naive triggers that temporary sudden spikes or dips due to reactions to news can activate a buy or a sale
    - These limits are good for sudden market changes that will persist and allow a user to protect themselves
- Because charts should be checked on a daily basis, there should be an easier way to obtain this information so users can make informed and timely investment decisions

User Benefits

- See today's signal for any tracked ticker at a glance
- Avoid having to configure charts, which can have a higher learning curve depending on the differences and nuances of each individual charting software
- Compare the computer read fetched from hard numbers against the AI (via VLM) side-by-side
    - View the original chart for reference and interpretation traceability
    - Deterministic check allows for a flag to be raised if there isn't agreement from the retrieved chart interpretation
- Drill into the per-indicator detail to understand why the system interprets BUY/ SELL/ HOLD
- Gamify the daily ritual of having to check the charts

Goals

- Expose the runDaily(ticker) pipeline through an internal API that returns the full dual-read payload of:
    - Fetched readings
    - VLM readings
    - Derived signals
    - Overall recommendation
    - Chart image
    - Metadata

- Build a FE that renders a report for the user-inputed ticker, containing:
    - Headline signal (BUY/ SELL/ HOLD)
    - Per-indicator breakdown (crossover, bars ago, qualified, derived signal)
        - Fetched data
        - Playwright-retrieved and VLM-interpreted chart
- Future: Support viewing multiple ticker results in a single session (dashboard or list)

Assumptions

- Keep the architecture is simple, this is a quick-reference tool in its MVP state that isn't meant to be a widely-distributed product with saved sessions
- The pipeline runs on-demand per ticker (not pre-computed or cached)
    - Each API call triggers a fresh TradingView capture and VLM analysis, so response times will be in the 10–20 second range per ticker
- The frontend is hosted on Vercel
- The Playwright capture + VLM analysis pipeline runs on a separate Render service with a persistent Chromium instance
- The Vercel API layer proxies requests to the pipeline service
    - This separation is necessary because browser automation requires a long-running process with a full Chromium binary, which exceeds serverless function size and timeout constraints
- Chart capture requires a headed or headful Chromium instance on the server,
- The VLM API key is available server-side via environment variable
- There is a 10-ticker evaluation set to form the ground truth
    - A human-provided interpretation of the chart and data form the ground truth
    - Timeseries data allows for regression testing of how effective these charts/ signals are if adhered to
    - Track and be alerted of model drift, processing time increases/ risks, UI changes that affect retrieval/ interpretation
- The existing runDaily return shape is stable enough to build against
- Indicator parameters
    - MACD: 8, 17, 9
    - Slow Stochastic: 14, 5
    - Simple MA: 10
- Window: ~6 months

User Stories		
Per-Indicator Breakdowns		
Title	User Story	Priority
Fetched Data	As a user, I want to see all relevant numerical data for the three indicators so I can verify the math and edge cases.	Essential
Chart Image	As a user, I want to see the image pulled from the indicator charting website so I can verify it was pulled and read correctly.	Essential
Image Interpretation	As a user, I want to see the model's individual component readings for the three indicators so I can verify correctness and account for edge cases.	Essential
Overall Suggestion		
Suggestion	As a user, I want to see the overall suggestion based on the "polled" results.	Essential
Agentic Disagreement Resolution	As a user, I want the agent to help parse and interpret disagreements between the VLM-interpretation and the fetched results.	Important
Rationale	As a user, I want the option to read why the overall suggestion is the way it is and if it differs, why the image-interpreted result and the calculated results might disagree.	Important

Comparison	As a user, I want to see how the retrieved and calculated results compare with the image-interpreted result so I can make my own informed decision.	Essential
Testing		
Ground Truth	As a data scientist, I want to maintain a set of 10 stock chart images that I've manually labeled for testing VLM performance.	Essential
Historical Testing	As a data scientist, I want to regression test whether entering or exiting a stock when the indicators agree to do so resulted in profits or not.	Nice to Have

User Story Details		
Fetched Data	As a user, I want to see all relevant numerical data for the three indicators so I can verify the math and edge cases.	
Area	**Description**	**Consideration/Flags**
Retrieved Numbers	☐ Crossover date + bars ago ☐ Bearish or Bullish	

Qualified	☐ Qualified boolean
	☐ Only for "Additional Details" on each indicator:
	    ☐ MACD: Bearish cross above 0/ Bullish cross below 0
	    ☐ Slow Stoch: Bearish cross above 80%/ Bullish cross below 20%	

	☐ SMA: Bearish cross on negative slope/ Bullish cross on positive slope	
Derived Signal	☐ Display derived signal based on data + qualified	
Image Interpretation	As a user, I want to see the model's individual component readings for the three indicators so I can verify correctness and account for edge cases.	
Area	**Description**	**Consideration**
Interpreted Points	☐ Crossover bars ago ☐ Bearish or Bullish	
Qualified	☐ Qualified boolean ☐ Only for "Additional Details" on each indicator, provide VLM rationale	
Derived Signal	☐ Display derived signal based on data + qualified	
Chart Image	As a user, I want to see the image pulled from the indicator charting website so I can verify it was pulled and read correctly.	
Area	**Description**	**Consideration/Flags**
Chart Image	☐ Should only be the designated region ☐ PII check ensures no identifiable information is in the screenshot ☐ Embed screenshot and place in center	
End Date (Optional)	☐ Optionally designate an end date after which documents will be no longer expected	
Suggestion	As a user, I want to see the overall suggestion based on the "polled" results.	

Area	Description	Consideration/Flags

Suggestion	☐ Display overall suggestion: Buy, Hold, Sell
	    ☐ Buy: All three indicators need to signal "Buy"
	    ☐ Sell: Two or more indicators need to signal "Sell"
	    ☐ Hold: All other cases, including "Neutral" signals	

Rationale	As a user, I want the option to read why the overall suggestion is the way it is and if it differs, why the image-interpreted result and the calculated results might disagree.	
Area	**Description**	**Consideration**
Provide Rationale	☐ Explain overall suggestion in plain English ☐ Provide details on difference between fetched and VLM, and any reconciliation if an agent performed this	
Agentic Disagreement Resolution	As a user, I want the agent to interpret disagreements between the VLM-interpretation and the fetched results.	
Area	**Description**	**Consideration**
De-conflict	☐ If there is a disagreement, delegate to an agent for reconciliation by considering the context	
Ground Truth	As a data scientist, I want to maintain a set of 10 stock chart images that I've manually labeled for testing VLM performance.	
Area	**Description**	**Consideration**

Labels	☐ Bars ago ☐ Qualified ☐ Bearish or Bullish ☐ Per Indicator: Buy/ Sell/ Neutral ☐ Overall: Buy/ Sell/ Hold	

Result Interpretation	☐ Neither the fetched values nor the VLM are treated as the ground truth, only the user labels are the ground truth
	☐ VLM Scoring:
	    ☐ Per indicator:
	        ☐ Bullish or Bearish
	        ☐ Qualified boolean
	        ☐ Bars ago
	        ☐ Buy/ sell/ neutral
	    ☐ Buy/ hold/ sell (polled result)
	☐ Fetched result scoring:
	    ☐ Per indicator:
	        ☐ Bullish or Bearish
	        ☐ Qualified boolean
	        ☐ Bars ago
	        ☐ Buy/ sell/ neutral
	    ☐ Buy/ hold/ sell (polled result)
	☐ Compare overall suggestion based on comparison; compare to overall ground truth for stock
	☐ Log run results over time to track model drift and fetch result errors	

Historical Testing	As a data scientist, I want to regression test whether entering or exiting a stock when the indicators agree to do so resulted in profits or not.	
Area	**Description**	**Consideration**
Regression Testing	☐ Done for each stock ticker ☐ Read signals from left to right, and: ☐ If Buy signal detected, log a	May be imprecise with prices as a graph must be read, or when historic indicator values and prices are no

	“purchase” of 1,000 shares at the price for that day ☐ If Sell signal detected and number of shares > 1,000, “sell” all shares at the price ☐ Use a calculate function to determine how much profit was made over all transactions for the stock	longer available for the period